

DPR.ai Frontend Modernization Execution Guide

Production-grade frontend architecture and AI execution roadmap

Phase 0 — Frontend Stabilization + Governance Foundation

Purpose: Stop frontend entropy before modernization begins.

Tasks Included
Analyze current frontend architecture and duplicate systems
Create frontend constitution and governance rules
Standardize token usage and spacing scale
Remove deprecated dashboards and dead files
Define folder ownership and AI coding standards
Perform visual consistency and dependency audits

Phase 1 — Foundation System (Tokens + Primitives)

Purpose: Build the reusable frontend operating system.

Tasks Included
Create semantic token architecture
Build typography scale and density system
Build Button, Badge, Spinner, Icon, Skeleton primitives
Define primitive contracts and accessibility rules
Create Storybook and Vitest testing setup
Remove all hardcoded colors and arbitrary Tailwind usage

Phase 2 — Application Shell + Navigation Architecture

Purpose: Create the universal application frame for all pages.

Tasks Included
Build PageShell, Sidebar, Topbar, MobileNav
Create role-aware navigation config system
Implement RoleContext and permission logic
Refactor giant sidebar monolith
Build responsive layout system
Add Playwright responsive layout tests

Phase 3 — Data Entry + Operational Workflow System

Purpose: Build operational interaction workflows.

Tasks Included
Build Input, Select, FormField, Textarea systems
Create reusable TanStack DataTable architecture
Implement OCR workflows and attendance systems
Define state ownership and mutation patterns
Refactor giant OCR and form components
Add workflow and validation tests

Phase 4 — Intelligence + Analytics Layer

Purpose: Build dashboards, analytics, KPI and reporting surfaces.

Tasks Included
Build KPI card system and analytics grids
Create ApexCharts theme system
Standardize chart wrappers and exports
Build alert feed and reporting workflows
Remove duplicated dashboards
Perform dashboard performance profiling

Phase 5 — Feedback + Polish + Experience Layer

Purpose: Make the frontend production-ready and governed.

Tasks Included
Build Toast, Modal, Tooltip and Overlay systems
Create empty/error/auth states
Implement keyboard accessibility and focus management
Build marketing and auth pages
Run full Playwright E2E testing
Perform final frontend governance audit

Global Frontend Governance Rules

- No hardcoded colors or spacing values
- No business logic inside primitive components
- No direct API calls inside UI primitives
- All pages must use PageShell
- All components require loading, empty, and error states
- All shared components must remain reusable and role-aware
- All frontend changes must preserve scalability across 40+ pages